

12 Functions

12.1 Functions

2. Suppose $A = \{a, b, c, d\}$, $B = \{2, 3, 4, 5, 6\}$ and

$$f = \{(a, 2), (b, 3), (c, 4), (d, 5)\}.$$

State the domain and range of f . Find $f(b)$ and $f(d)$.

The domain of f is A and the range of f is $\{2, 3, 4, 5\}$.

$f(b) = 3$ and $f(d) = 5$.

4. There are eight different functions $f : \{a, b, c\} \rightarrow \{0, 1\}$. List them all. Diagrams will suffice.

$$f_1 = \{(a, 0), (b, 0), (c, 0)\}$$

$$f_2 = \{(a, 0), (b, 0), (c, 1)\}$$

$$f_3 = \{(a, 0), (b, 1), (c, 0)\}$$

$$f_4 = \{(a, 0), (b, 1), (c, 1)\}$$

$$f_5 = \{(a, 1), (b, 0), (c, 0)\}$$

$$f_6 = \{(a, 1), (b, 0), (c, 1)\}$$

$$f_7 = \{(a, 1), (b, 1), (c, 0)\}$$

$$f_8 = \{(a, 1), (b, 1), (c, 1)\}$$

6. Suppose $f : \mathbb{Z} \rightarrow \mathbb{Z}$ is defined as $f = \{(x, 4x + 5) : x \in \mathbb{Z}\}$. State the domain, codomain and range of f . Find $f(10)$.

The domain and codomain of f are both $\mathbb{Z}$. The range is the set of values $4x + 5$ for $x \in \mathbb{Z}$, that is, the integers that are one more than a multiple of 4:

$$\{4x + 5 : x \in \mathbb{Z}\} = \{\dots, -7, -3, 1, 5, 9, 13, \dots\}.$$

8. Consider the set $f = \{(x, y) \in \mathbb{Z} \times \mathbb{Z} : x + 3y = 4\}$. Is this a function from $\mathbb{Z}$ to $\mathbb{Z}$? Explain.

No. For f to be a function from $\mathbb{Z}$ to $\mathbb{Z}$, every integer must occur as the first coordinate of some pair in f . But $x = 0$ does not: that would require $3y = 4$, which has no integer solution y . So 0 is not the first coordinate of any element of f , and f is not a function from $\mathbb{Z}$ to $\mathbb{Z}$.

10. Consider the set $f = \{(x^3, x) : x \in \mathbb{R}\}$. Is this a function from $\mathbb{R}$ to $\mathbb{R}$? Explain.

Yes, this is a function $f : \mathbb{R} \rightarrow \mathbb{R}$. Every real number k is the cube of exactly one real number, namely $x = \sqrt[3]{k}$, so each $k \in \mathbb{R}$ occurs exactly once as the first coordinate of an ordered pair in f .

12. Is the set $\theta = \{((x, y), (3y, 2x, x + y)) : x, y \in \mathbb{R}\}$ a function? If so, what is its domain, codomain and range?

Yes. Every first coordinate is a pair $(x, y) \in \mathbb{R}^2$, and each such pair occurs exactly once, so θ is a function $\theta : \mathbb{R}^2 \rightarrow \mathbb{R}^3$ given by $\theta(x, y) = (3y, 2x, x + y)$. Its domain is $\mathbb{R}^2$ and its codomain is $\mathbb{R}^3$.

For the range, a triple $(a, b, c) \in \mathbb{R}^3$ equals $\theta(x, y)$ exactly when $a = 3y$, $b = 2x$ and $c = x + y$, that is, when $y = \frac{a}{3}$, $x = \frac{b}{2}$ and $c = \frac{b}{2} + \frac{a}{3}$. Conversely, given any (a, b, c) with $c = \frac{a}{3} + \frac{b}{2}$, taking $x = \frac{b}{2}$ and $y = \frac{a}{3}$ gives $\theta(x, y) = (a, b, c)$. Therefore the range is

$$\{(a, b, c) \in \mathbb{R}^3 : 2a + 3b = 6c\},$$

a plane through the origin in $\mathbb{R}^3$.

12.2 Injective and Surjective Functions

2. Consider the logarithm function $\ln : (0, \infty) \rightarrow \mathbb{R}$. Decide whether this function is injective and whether it is surjective.

This function is bijective. Observe that, by properties of the exponential function, $e^{\ln x} = x$ for every $x \in (0, \infty)$, and $\ln e^b = b$ for every $b \in \mathbb{R}$.

First we show $\ln$ is injective. Suppose $\ln m = \ln n$ for some $m, n \in (0, \infty)$. Applying the exponential function to both sides gives $e^{\ln m} = e^{\ln n}$, that is, $m = n$. Thus $\ln$ is injective.

Next we show $\ln$ is surjective. Take any $b \in \mathbb{R}$, and let $a = e^b$. Then $a \in (0, \infty)$, since $e^b > 0$ for every real b , and $\ln a = \ln e^b = b$. So every $b \in \mathbb{R}$ equals $\ln a$ for some a in the domain, and $\ln$ is surjective.

Therefore $\ln$ is bijective since it is both injective and surjective on this domain.

4. A function $f : \mathbb{Z} \rightarrow \mathbb{Z} \times \mathbb{Z}$ is defined as $f(n) = (2n, n + 3)$. Verify whether this function is injective and whether it is surjective.

This function is injective but not surjective.

First we show f is injective. Suppose $f(m) = f(n)$ for some $m, n \in \mathbb{Z}$. This means $(2m, m + 3) = (2n, n + 3)$, thus $2m = 2n$. Dividing this equation by two gives $m = n$. Thus f is injective.

Next we show f is not surjective. Consider $(1, 0) \in \mathbb{Z} \times \mathbb{Z}$. If $f(n) = (1, 0)$ for some $n \in \mathbb{Z}$, then equating the first coordinates gives $2n = 1$, which has no solution $n \in \mathbb{Z}$. So no integer maps to $(1, 0)$, and f is not surjective.

6. A function $f : \mathbb{Z} \times \mathbb{Z} \rightarrow \mathbb{Z}$ is defined as $f(m, n) = 3n - 4m$. Verify whether this function is injective and whether it is surjective.

This function is surjective but not injective.

First we show f is not injective. Observe that $f(0, 0) = 3(0) - 4(0) = 0$ and $f(3, 4) = 3(4) - 4(3) = 12 - 12 = 0$. Thus $f(0, 0) = f(3, 4)$ even though $(0, 0) \neq (3, 4)$, so f is not injective.

Next we show f is surjective. Take any $b \in \mathbb{Z}$. Then $(-b, -b) \in \mathbb{Z} \times \mathbb{Z}$, and $f(-b, -b) = 3(-b) - 4(-b) = -3b + 4b = b$. So every $b \in \mathbb{Z}$ equals $f(m, n)$ for some (m, n) in the domain, and f is surjective.

8. A function $f : \mathbb{Z} \times \mathbb{Z} \rightarrow \mathbb{Z} \times \mathbb{Z}$ is defined as $f(m, n) = (m + n, 2m + n)$. Verify whether this function is injective and whether it is surjective.

This function is both injective and surjective, hence bijective.

First we show f is injective. Suppose $f(m, n) = f(k, \ell)$ for some $(m, n), (k, \ell) \in \mathbb{Z} \times \mathbb{Z}$. This means $(m + n, 2m + n) = (k + \ell, 2k + \ell)$. Equating corresponding coordinates gives the two equations $m + n = k + \ell$ and $2m + n = 2k + \ell$. Subtracting the first from the second yields $m = k$, and substituting this back into the first gives $n = \ell$. Therefore $(m, n) = (k, \ell)$, so f is injective.

Next we show f is surjective. Take any $(a, b) \in \mathbb{Z} \times \mathbb{Z}$, and let $m = b - a$ and $n = 2a - b$. So $(m, n) \in \mathbb{Z} \times \mathbb{Z}$, and $f(m, n) = ((b - a) + (2a - b), 2(b - a) + (2a - b)) = (a, b)$. So every $(a, b) \in \mathbb{Z} \times \mathbb{Z}$ equals $f(m, n)$ for some (m, n) in the domain, and f is surjective.

Being both injective and surjective, f is bijective.

10. Prove the function $f : \mathbb{R} - \{1\} \rightarrow \mathbb{R} - \{1\}$ defined by $f(x) = \left(\frac{x+1}{x-1}\right)^3$ is bijective.

Proof. First we show f is injective. Suppose $f(x) = f(y)$ for some $x, y \in \mathbb{R} - \{1\}$, that is,

$$\left(\frac{x+1}{x-1}\right)^3 = \left(\frac{y+1}{y-1}\right)^3.$$

Every real number has exactly one real cube root, so taking cube roots of both sides gives

$$\frac{x+1}{x-1} = \frac{y+1}{y-1}.$$

Since $x \neq 1$ and $y \neq 1$, both denominators are nonzero, so we may cross-multiply to get $(x+1)(y-1) = (y+1)(x-1)$. Expanding both sides,

$$xy - x + y - 1 = xy - y + x - 1.$$

Cancelling xy and -1 leaves $-x + y = -y + x$, that is, $2y = 2x$, so $x = y$. Thus f is injective.

Next we show f is surjective. Take any $b \in \mathbb{R} - \{1\}$ and let $c = \sqrt[3]{b}$. Note that $c \neq 1$, for otherwise $b = c^3 = 1$. Now let

$$x = \frac{c+1}{c-1},$$

which is defined because $c \neq 1$. Also $x \neq 1$, since $x = 1$ would give $c+1 = c-1$, which is impossible. Hence $x \in \mathbb{R} - \{1\}$. Observe that

$$x+1 = \frac{c+1}{c-1} + 1 = \frac{2c}{c-1} \quad \text{and} \quad x-1 = \frac{c+1}{c-1} - 1 = \frac{2}{c-1},$$

so $\frac{x+1}{x-1} = \frac{2c}{2} = c$, and therefore

$$f(x) = \left(\frac{x+1}{x-1}\right)^3 = c^3 = b.$$

So every $b \in \mathbb{R} - \{1\}$ equals $f(x)$ for some x in the domain, and f is surjective.

Being both injective and surjective, f is bijective. □

12. Consider the function $\theta : \{0, 1\} \times \mathbb{N} \rightarrow \mathbb{Z}$ defined as $\theta(a, b) = a - 2ab + b$. Is θ injective? Is it surjective? Bijective? Explain.

This function is bijective. Since the first coordinate a can only be 0 or 1, we compute the two cases separately: $\theta(0, b) = 0 - 2(0)b + b = b$ and $\theta(1, b) = 1 - 2b + b = 1 - b$. As b ranges over $\mathbb{N}$, the first line produces exactly the integers $1, 2, 3, \dots$, and the second produces exactly $0, -1, -2, \dots$, that is, the integers less than or equal to 0.

14. Consider the function $\theta : \mathcal{P}(\mathbb{Z}) \rightarrow \mathcal{P}(\mathbb{Z})$ defined as $\theta(X) = \overline{\overline{X}}$. Is θ injective? Is it surjective? Bijective? Explain.

This function is a bijection, to show this, note that $\overline{\overline{\overline{X}}} = X$ for every $X \subseteq \mathbb{Z}$.

To see that θ is injective, suppose $\theta(X) = \theta(Y)$ for some $X, Y \in \mathcal{P}(\mathbb{Z})$, that is, $\overline{X} = \overline{Y}$. Taking complements of both sides gives $\overline{\overline{X}} = \overline{\overline{Y}}$, that is, $X = Y$. Thus θ is bijective.

To see that θ is surjective, take any $Y \in \mathcal{P}(\mathbb{Z})$. Then $\overline{Y}$ is also a subset of $\mathbb{Z}$, so $\overline{Y} \in \mathcal{P}(\mathbb{Z})$, and $\theta(\overline{Y}) = \overline{\overline{Y}} = Y$.

Being both injective and surjective, θ is bijective.

16. This question concerns functions $f : \{A, B, C, D, E\} \rightarrow \{1, 2, 3, 4, 5, 6, 7\}$. How many such functions are there? How many of these functions are injective? How many are surjective? How many are bijective?

The domain has 5 elements and the codomain has 7.

A function can be determined by choosing a value in $\{1, 2, \dots, 7\}$ for each of the 5 elements of the domain, so by the multiplication principle there are $7^5 = 168074$ such functions.

Now, the 5 values must be distinct, so there are 7 choices for the image of A , then 6 for B , then 5, 4 and 3. This gives $7 \cdot 6 \cdot 5 \cdot 4 \cdot 3 = 2520$ injective functions.

There are no surjective functions as this would require every element in the codomain to be $f(x)$ for some x in the domain, but the domain supplies only 5 values.

Because none of the functions are surjective, none of the functions are bijective.

12.3 The Pigeonhole Principle

2. Prove that if a is a natural number, then there exist two unequal natural numbers k and ℓ for which $a^k - a^\ell$ is divisible by 10.

4. Consider a square whose side-length is one unit. Select any five points from inside this square. Prove that at least two of these points are within $\frac{\sqrt{2}}{2}$ units of each other.

6. Given a sphere S , a *great circle* of S is the intersection of S with a plane through its center. Every great circle divides S into two parts. A *hemisphere* is the union of the great circle and one of these two parts. Prove that if five points are placed arbitrarily on S , then there is a hemisphere that contains four of them.

12.4 Composition

2. Suppose $A = \{1, 2, 3, 4\}$, $B = \{0, 1, 2\}$, $C = \{1, 2, 3\}$. Let $f : A \rightarrow B$ be

$$f = \{(1, 0), (2, 1), (3, 2), (4, 0)\},$$

and $g : B \rightarrow C$ be $g = \{(0, 1), (1, 1), (2, 3)\}$. Find $g \circ f$.

4. Suppose $A = \{a, b, c\}$. Let $f : A \rightarrow A$ be the function

$$f = \{(a, c), (b, c), (c, c)\},$$

and let $g : A \rightarrow A$ be the function $g = \{(a, a), (b, b), (c, a)\}$. Find $g \circ f$ and $f \circ g$.

6. Consider the functions $f, g : \mathbb{R} \rightarrow \mathbb{R}$ defined as $f(x) = \frac{1}{x^2 + 1}$ and $g(x) = 3x + 2$. Find the formulas for $g \circ f$ and $f \circ g$.

8. Consider the functions $f, g : \mathbb{Z} \times \mathbb{Z} \rightarrow \mathbb{Z} \times \mathbb{Z}$ defined as $f(m, n) = (3m - 4n, 2m + n)$ and $g(m, n) = (5m + n, m)$. Find the formulas for $g \circ f$ and $f \circ g$.

10. Consider the function $f : \mathbb{R}^2 \rightarrow \mathbb{R}^2$ defined by the formula $f(x, y) = (xy, x^3)$. Find a formula for $f \circ f$.

12.5 Inverse Functions

2. In Exercise 9 of Section 12.2 you proved that $f : \mathbb{R} - \{2\} \rightarrow \mathbb{R} - \{5\}$ defined by $f(x) = \frac{5x + 1}{x - 2}$ is bijective. Now find its inverse.

4. The function $f : \mathbb{R} \rightarrow (0, \infty)$ defined as $f(x) = e^{x^3 + 1}$ is bijective. Find its inverse.

6. The function $f : \mathbb{Z} \times \mathbb{Z} \rightarrow \mathbb{Z} \times \mathbb{Z}$ defined by the formula $f(m, n) = (5m + 4n, 4m + 3n)$ is bijective. Find its inverse.

8. Is the function $\theta : \mathcal{P}(\mathbb{Z}) \rightarrow \mathcal{P}(\mathbb{Z})$ defined as $\theta(X) = \overline{X}$ bijective? If so, what is its inverse?

10. Consider $f : \mathbb{N} \rightarrow \mathbb{Z}$ defined as $f(n) = \frac{(-1)^n(2n - 1) + 1}{4}$. This function is bijective by Exercise 18 in Section 12.2. Find its inverse.

12.6 Image and Preimage

2. Consider the function $f : \{1, 2, 3, 4, 5, 6, 7\} \rightarrow \{0, 1, 2, 3, 4, 5, 6, 7, 8, 9\}$ given as

$$f = \{(1, 3), (2, 8), (3, 3), (4, 1), (5, 2), (6, 4), (7, 6)\}.$$

Find:

$$f(\{1, 2, 3\}), \quad f(\{4, 5, 6, 7\}), \quad f(\emptyset), \quad f^{-1}(\{0, 5, 9\}) \quad \text{and} \quad f^{-1}(\{0, 3, 5, 9\}).$$

4. This problem concerns functions

$$f : \{1, 2, 3, 4, 5, 6, 7, 8\} \rightarrow \{0, 1, 2, 3, 4, 5, 6\}.$$

How many such functions have the property that $|f^{-1}(\{2\})| = 4$?

6. Given a function $f : A \rightarrow B$ and a subset $Y \subseteq B$, is $f(f^{-1}(Y)) = Y$ always true? Prove or give a counterexample.

8. Given a function $f : A \rightarrow B$ and subsets $W, X \subseteq A$, then $f(W \cap X) = f(W) \cap f(X)$ is false in general. Produce a counterexample.

10. Given $f : A \rightarrow B$ and subsets $Y, Z \subseteq B$, prove $f^{-1}(Y \cap Z) = f^{-1}(Y) \cap f^{-1}(Z)$.

12. Consider $f : A \rightarrow B$. Prove that f is injective if and only if $X = f^{-1}(f(X))$ for all $X \subseteq A$. Prove that f is surjective if and only if $f(f^{-1}(Y)) = Y$ for all $Y \subseteq B$.

14. Let $f : A \rightarrow B$ be a function, and $Y \subseteq B$. Prove or disprove: $f^{-1}(f(f^{-1}(Y))) = f^{-1}(Y)$.